World-Formation Formal Core v1.0 A Minimal Formal Theory of Bounded Observers, Declaration, Purpose, Trace, Residual, Latching, and Revision

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Abstract

World-Formation Formal Core v1.0 develops a minimal formal architecture for bounded observers capable of forming, maintaining, auditing, and revising operational worlds.

The theory begins without assuming a particular physical substrate or higher mathematical geometry. Its primitive functional roles are **Observer, Declaration, Purpose, Gate, Trace, Filtration, Residual, Latching, and Revision**. These components are explicitly separated from optional mathematical extensions such as octonions, quaternionic subalgebras, $G_2/SO(4)$, symplectic geometry, complex structures, Clifford constructions, and bundle geometry. The source development likewise separates these layers and prohibits later interpretive structures from retrospectively establishing the Core. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

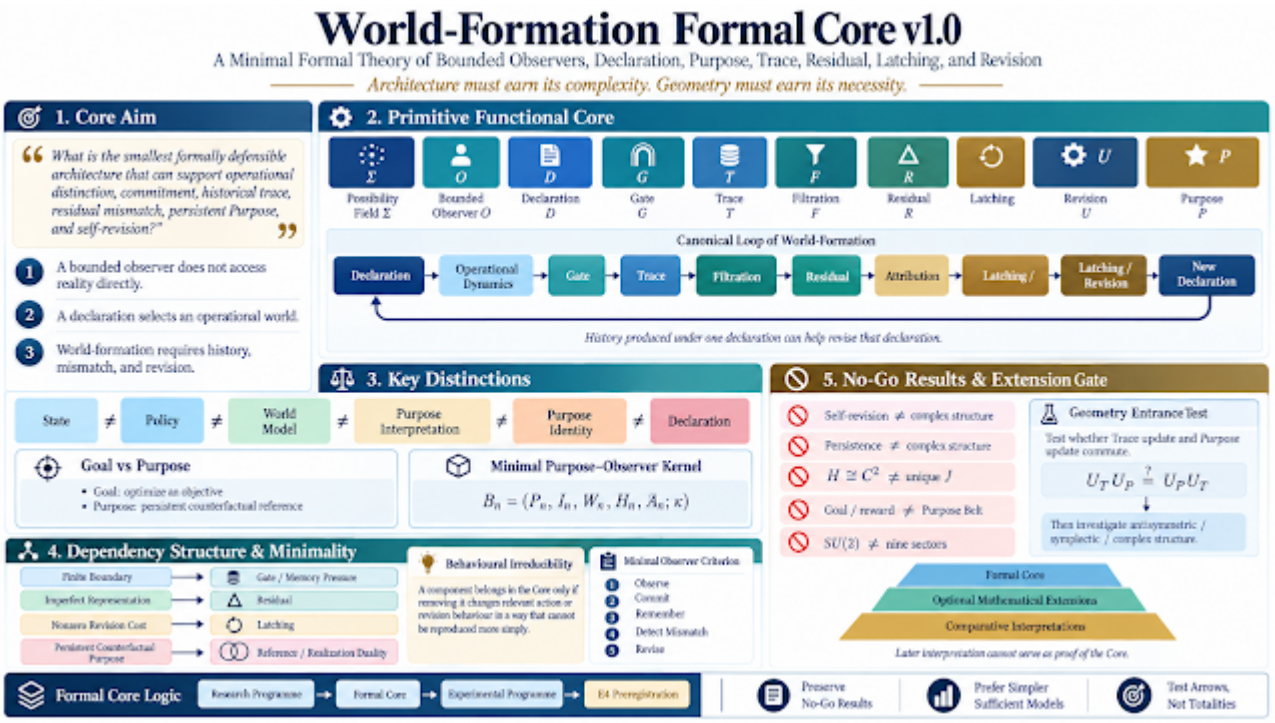
A bounded observer operates through a Declaration D that determines an operational world W_D. Observations do not automatically become history: a Gate G determines commitment, producing Trace T and an accumulating Filtration F. Because the declaration is finite and potentially incomplete, Residual R records mismatch between the current operational world and encountered evidence. Latching introduces historical resistance to arbitrary revision, while Revision U permits the system to modify policy, world model, Purpose interpretation, Purpose identity, or Declaration itself.

The formal theory further distinguishes **Goal** from **Purpose**. Purpose is treated as a persistent counterfactual reference that remains distinguishable from realised history and from its current interpretation. This makes possible a self-referential system in which the history generated under one declaration can later participate in revising the declaration through which that history became meaningful.

The formalism deliberately preserves several negative results. Persistence and self-revision can exist entirely in real-valued dynamics and therefore do not imply complex structure. Goal optimisation does not imply a Purpose Belt. The equivalence $\mathbb{H} \cong \mathbb{C}^2$ does not select a unique complex structure. Deeper geometry must therefore enter only after the functional Core establishes a phenomenon that requires it. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

The central methodological criterion is behavioural minimality:

> **A proposed component belongs in the Core only if removing it changes relevant action or revision behaviour in a way that cannot be reproduced by a simpler state representation.**



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1. Scope

The Science of World-Formation asks how an operational world becomes available to a bounded observer.

The Formal Core addresses a narrower problem:

> **What is the smallest formally defensible architecture that can support operational distinction, commitment, historical trace, residual mismatch, persistent Purpose, and self-revision?**

The aim is not to maximise metaphysical scope.

It is to minimise assumptions while preserving the distinctive phenomenon under study.

The core research object is therefore not a universe in itself, but a recursive relation:

Observer ↔ Declared World ↔ Historical Trace ↔ Residual ↔ Revision. (1.1)

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2. Epistemic Status Ledger

Every object or statement in the Formal Core is assigned one of the following statuses.

Status	Meaning
[P] Primitive	Functional role not further derived within the current Core
[A] Assumption	Explicit condition introduced for a particular result
[K] Known Mathematics	Imported established mathematical result
[D] Derived Result	Result following from stated assumptions
[C] Construction	A useful formal realisation not claimed to be unique

Status	Meaning
[H] Hypothesis	Claim awaiting proof or experiment
[NG] No-Go Result	A demonstrated failure of a weaker condition to imply a stronger one
[S] Superseded	Earlier formulation replaced or downgraded
[I] Interpretation	Comparative or philosophical reading outside the Formal Core

The source development explicitly calls for a ledger of Primitive, Assumption, Known Mathematics, Derived Result, Construction, Hypothesis, No-Go, and Superseded claims before rebuilding the dependency graph. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

The present paper follows that discipline.

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3. Possibility Domain

Definition 3.1 — Possibility Field [P]

Let Σ denote a domain of possibilities not yet fully organised under a specific operational declaration.

No particular ontology is assumed.

Σ may represent:

- possible environmental states;
- possible observations;
- candidate models;
- semantic alternatives;
- possible actions;
- possible relations;
- or a more abstract pre-declared carrier.

The only required condition is that a bounded observer does not possess unrestricted operational access to all of Σ.

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4. Bounded Observer

Definition 4.1 — Observer [P]

A bounded observer O is a system whose access to Σ is limited by finite representation, finite observation, finite memory, finite action, finite computation, or finite revision capacity.

Schematically:

$O(\Sigma) \neq \Sigma.$ (4.1)

Boundedness therefore creates a distinction between:

what is possible, (4.2)

and

what is operationally admitted. (4.3)

This distinction motivates Declaration.

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5. Declaration

Definition 5.1 — Declaration [P]

A Declaration D is a rule, structure, or protocol that maps a possibility domain into an operationally admitted world:

$D: \Sigma \to W_D$. (5.1)

W_D is called the **declared world** under D.

Declaration may determine:

- admitted variables;
- distinctions;
- relations;
- ontology;
- representation;
- evidence rules;
- permissible operations;
- action classes;
- observational interfaces;
- model classes;
- revision rules.

Thus:

Declaration = operational world selection. (5.2)

The source development explicitly elevates Declaration as model/world selection rather than merely a descriptive statement. $\mathbb{O} \to G_2/SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

Definition 5.2 — Declared State Space [C]

Given D, let X_D denote the state space available within the declared world.

Then:

$x_t \in X_D$. (5.3)

Different declarations need not produce equivalent state spaces:

$X_{D_1} \ncong X_{D_2}$ in general. (5.4)

A declaration revision may therefore change not only parameter values but the representational space in which parameters are defined.

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6. Ordinary Learning and Declaration Revision

Let θ denote parameters inside a fixed declaration.

Ordinary learning has the form:

$\theta_n \rightarrow \theta_{n+1}$, with D fixed. (6.1)

Declaration revision instead has the form:

$D_n \rightarrow D_{n+1}$. (6.2)

The source explicitly distinguishes parameter change within D_n from PORE-style revision in which D_n itself becomes D_{n+1}, meaning that representation or model class changes. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

This motivates a hierarchy:

State revision ≠ Policy revision ≠ Purpose revision ≠ Declaration revision. (6.3)

The distinction is functional rather than merely terminological.

Different failures may require different revision levels.

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7. Gate

Definition 7.1 — Gate [P]

A Gate G determines whether a candidate event, observation, inference, or decision becomes historically committed.

Let z_n denote a candidate observation.

Then:

$g_n = G(z_n \mid D_n, P_n, F_n)$. (7.1)

In the simplest binary construction:

$g_n \in \{0,1\}$. (7.2)

The Gate separates:

candidate possibility ≠ committed historical event. (7.3)

The precise implementation may be probabilistic, thresholded, categorical, or continuous.

The primitive role is commitment.

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8. Trace

Definition 8.1 — Trace [P]

A Trace T_n is a committed historical record generated when a candidate outcome passes the Gate.

A simple construction is:

$T_n = g_n \odot z_n.$ (8.1)

The exact algebra is optional.

The defining property is:

Trace = observation after commitment. (8.2)

Trace matters because committed history can later constrain interpretation and revision.

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9. Filtration

Definition 9.1 — Filtration [P]

Let F_n denote the accumulated history available to the observer after n committed updates.

Then:

$F_0 \subseteq F_1 \subseteq F_2 \subseteq \dots$ (9.1)

A filtration is therefore not merely storage.

It is ordered disclosure.

The source architecture already contains Trace, Filtration, Adaptive Policy, and Latching on the observer side, allowing later Purpose structures to reuse the existing historical machinery rather than duplicate it. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

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10. Residual

Definition 10.1 — Residual [P]

Residual R is the mismatch between the current declared world and what the observer cannot adequately absorb, predict, reconstruct, or reconcile.

Abstractly:

$R_n = E(D_n, P_n, F_n).$ (10.1)

No linear structure is required.

If a projection-like construction exists, one may write:

$r_D(x) = x - D(x).$ (10.2)

But equation (10.2) is only a construction, not a primitive definition.

Definition 10.2 — Residual Magnitude [C]

Let $\mathscr{R}_n \geq 0$ measure total mismatch.

A simple construction is:

$\mathscr{R}_n = \Sigma_{k \le n} w_k \|e_k\|^2$. (10.3)

This answers:

> How much mismatch has accumulated?

Definition 10.3 — Directional Residual [C]

Let $\mathcal{G}_n$ measure whether residuals accumulate coherently in a particular direction.

A simple construction is:

$\mathcal{G}_n = \Sigma_{k \le n} a_k e_k$. (10.4)

For declaration space $\mathscr{M}$_D, one may instead write:

F_R ∈ T_D$\mathscr{M}$_D. (10.5)

This answers:

> Does the accumulated mismatch systematically favour a particular revision?

The source development explicitly distinguishes residual magnitude from directional residual. Large total error with near-zero direction may reflect noise, whereas individually small but consistently directed errors may reveal structural bias. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

Thus:

large $\mathscr{R} + \mathcal{G} \approx 0$ may indicate noise. (10.6)

small $\mathscr{R}$ per event + persistent $\mathcal{G}$ may indicate structural mismatch. (10.7)

This distinction is a hypothesis to be experimentally tested rather than assumed universally.

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11. Latching

Definition 11.1 — Latching [P]

Latching is the persistence of historical or structural commitment under evidence insufficient to justify revision.

Let κ_ℓ be the switching cost associated with revision level ℓ.

A generic rule is:

revise at level ℓ only if ΔL_ℓ > κ_ℓ. (11.1)

Otherwise:

Latch. (11.2)

Latching therefore creates a finite persistence regime between unrestricted plasticity and

complete rigidity.

The source explicitly argues that Purpose should be revisable but not casually revisable, and proposes finite level-dependent revision costs. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

Definition 11.2 — Hierarchical Revision Cost [C]

A candidate ordering is:

κ_state < κ_policy < κ_world < κ_interpretation < κ_Purpose. (11.3)

This ordering is not a Core theorem.

It is a testable construction expressing the intuition that increasingly structural commitments should normally require increasingly strong evidence to change.

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12. Revision

Definition 12.1 — Revision Operator [P]

Revision U changes some component of the observer–world architecture in response to residual evidence.

Write:

$\Omega_{n+1} = U(\Omega_n, R_n)$. (12.1)

Revision need not always alter Declaration.

It may occur at several levels.

Definition 12.2 — Revision Level [C]

Let:

$\ell \in \{x, \pi, W, I, P, D\}$. (12.2)

where:

x = state estimate,
π = policy,
W = world model,
I = Purpose interpretation,
P = Purpose identity,
D = structural Declaration.

The exact taxonomy may later be reduced.

Its immediate purpose is to prevent all discrepancy from collapsing into generic parameter update.

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13. Goal and Purpose

Definition 13.1 — Goal [P]

A Goal may be represented as an objective, target, or loss function.

For example:

$\min_x L(x)$. (13.1)

A goal may be optimised without preserving any explicit distinction between its original identity and its current interpretation.

Definition 13.2 — Purpose [P]

Purpose is a persistent counterfactual reference against which realised history and current interpretation can be judged.

Purpose therefore maintains the distinction:

reference ≠ realisation. (13.2)

The source explicitly distinguishes Purpose from scalar reward and from system prompts. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

The minimal functional question is not:

> What reward should be maximised?

It is:

> What remains invariant enough that the system can judge whether its own reinterpretations still constitute a legitimate continuation of the same commitment?

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14. Purpose Identity and Interpretation

Definition 14.1 — Purpose Identity [P]

Let P_n denote the persistent identity of Purpose.

P_n is not required to remain absolutely constant.

It must instead remain sufficiently stable that change can be distinguished from continuation.

Definition 14.2 — Purpose Interpretation [P]

Let I_n denote the current operational interpretation of P_n under world model W_n.

Thus:

$I_n = \text{Interpret}(P_n, W_n, F_n)$. (14.1)

An ontology shift may therefore produce:

$W_n \rightarrow W_{n+1}, \quad (14.2)$

$I_n \rightarrow I_{n+1}, \quad (14.3)$

while still preserving:

$P_{n+1} \approx P_n. \quad (14.4)$

This allows reinterpretation without silent Purpose replacement.

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15. Revision Attribution

Definition 15.1 — Attribution [P]

Residual is insufficient by itself.

A discrepancy may indicate:

- policy failure;
- world-model failure;
- Purpose-interpretation failure;
- Purpose-identity failure;
- structural Declaration failure.

Let:

$A(R_n) = P(\ell \mid R_n, F_n, D_n, P_n). \quad (15.1)$

Revision then becomes conditional on diagnosis.

For example:

$\ell = \pi \Rightarrow \pi_n \rightarrow \pi_{n+1}. \quad (15.2)$

$\ell = W \Rightarrow W_n \rightarrow W_{n+1}. \quad (15.3)$

$\ell = I \Rightarrow I_n \rightarrow I_{n+1}. \quad (15.4)$

$\ell = P \Rightarrow P_n \rightarrow P_{n+1}. \quad (15.5)$

$\ell = D \Rightarrow D_n \rightarrow D_{n+1}. \quad (15.6)$

The source identifies Revision Attribution as a central functional addition. Without different diagnoses producing different revision classes, Purpose, interpretation, and world model risk becoming different names for ordinary parameters. $\mathbb{O} \rightarrow G_2_SO(4) \rightarrow \mathbb{H} \rightarrow \mathbb{C}^2$ 成界過程初探 1...

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16. Minimal Purpose–Observer Kernel

The initial Purpose Belt concept can be reduced substantially.

The source development argues that the existing self-referential observer architecture already

supplies:

Trace + Filtration + Adaptive Policy + Latching. (16.1)

The Purpose-specific layer then needs only to add:

Purpose Identity + Interpretation + Revision Attribution + Purpose-level Latching. (16.2)

This yields:

Self-Referential Observer + Purpose Interpretation Layer + Revision Governor. (16.3)

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A minimal candidate Purpose state is therefore:

$B_n = (P_n, I_n, W_n, H_n, A_n; \kappa)$. (16.4)

where H_n denotes sufficient historical state.

The full historical trajectories need not be stored if sufficient statistics preserve relevant behaviour. The source explicitly proposes this compression. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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17. Behavioural Irreducibility

Definition 17.1 — Irreducible Component

Let c be a proposed component of architecture B.

Remove c and obtain B \ c.

Component c is behaviourally irreducible over task family $\mathscr{H}$ only if no lower-complexity representation Z reproduces all relevant behaviour.

Formally, c is irreducible if there exists $h \in \mathscr{H}$ such that, for every admissible lower-complexity Z:

$P(A_{t:t+k}, U_{t:t+k} \mid Z, h) \neq P(A_{t:t+k}, U_{t:t+k} \mid B, h)$. (17.1)

If a smaller Z reproduces both action and revision behaviour:

c is bookkeeping relative to $\mathscr{H}$. (17.2)

The source proposes essentially this test as the cleanest formal criterion for Purpose Belt minimality. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

This principle extends to the entire Formal Core.

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18. Canonical World-Formation State

Construction 18.1 — Core State [C]

Define:

$\Omega_n = (D_n, P_n, I_n, W_n, x_n, F_n, R_n)$. (18.1)

Not every implementation must explicitly store every component.

Equation (18.1) is a bookkeeping construction for formal analysis.

The canonical update proceeds as follows.

Operational dynamics:

$x_{n+1} = \Phi_D_n,P_n(x_n, u_n, \xi_n)$. (18.2)

Candidate observation:

$z_n = H_D_n(x_{n+1})$. (18.3)

Commitment gate:

$g_n = G(z_n \mid D_n, P_n, I_n, F_n)$. (18.4)

Trace:

$T_n = \text{Commit}(g_n, z_n)$. (18.5)

Filtration update:

$F_{n+1} = F_n \vee T_n$. (18.6)

Residual evaluation:

$R_{n+1} = E(D_n, P_n, I_n, W_n, F_{n+1})$. (18.7)

Revision attribution:

$\ell_{n+1} = A(R_{n+1} \mid D_n, P_n, I_n, W_n, F_{n+1})$. (18.8)

Candidate revision:

$\hat{\Omega}_{n+1} = U^{\wedge}(\ell_{n+1})(\Omega_n, R_{n+1})$. (18.9)

Latching decision:

$\Omega_{n+1} = \text{LatchOrAccept}(\Omega_n, \hat{\Omega}_{n+1}; \kappa_\ell)$. (18.10)

These equations are a **construction introduced by this paper** to compact the source architecture. They are not claimed to be the only formalisation.

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19. The World-Formation Loop

The complete cycle can be compressed as:

Declaration → Operational Dynamics → Gate → Trace → Filtration → Residual → Attribution → Latching / Revision → New Declaration. (19.1)

Purpose runs through the cycle as a persistent counterfactual reference.

The distinctive recursive property is:

history generated under D_n may contribute to the transition $D_n \to D_{n+1}$. (19.2)

Thus the system does not merely update its state inside a fixed world.

It can revise the structure through which states, observations, and failures become intelligible.

This is the minimal formal meaning of **world-formation**.

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20. Assumption Dependency Graph

The current theory should not be read as one uninterrupted derivation.

It is a dependency graph.

The source explicitly recommends reconstructing the theory in this way. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1...

A1 — Finite Persistent Boundary

A bounded observer has finite persistent operational closure.

Candidate consequence:

A1 → Gate / memory pressure. (20.1)

A2 — Imperfect Representation

The current Declaration does not perfectly represent all relevant possibilities.

Candidate consequence:

A2 → Residual. (20.2)

A3 — Nonzero Revision Cost

Revision has nonzero cost.

Candidate consequence:

A3 → Latching. (20.3)

A4 — Persistent Counterfactual Purpose

Purpose persists sufficiently across time to remain distinguishable from realised history.

Candidate consequence:

A4 → Reference / realisation duality. (20.4)

A5 — Accountable Orientation

Revision carries ordered or directed accountability, not merely unsigned deviation.

Candidate consequence:

A5 → candidate antisymmetric relational structure. (20.5)

This is not yet equivalent to a symplectic form.

A6 — Nondegeneracy

If an independently obtained antisymmetric structure remains nondegenerate after appropriate quotienting:

A5 + A6 → candidate symplectic structure. (20.6)

A7 — Positive Purpose Metric

If there exists a positive Purpose metric g_P:

$g_P > 0$. (20.7)

then, together with an appropriate nondegenerate antisymmetric structure, one may investigate compatible complexification.

A8 — Real Dimension Four

If a relevant operational space V has:

$\dim_{\mathbb{R}}(V) = 4$, (20.8)

and a compatible complex structure J exists, then:

$\dim_{\mathbb{C}}(V) = 2$. (20.9)

Dimension alone does not supply J.

A9 — Quaternionic Compatibility

If V carries quaternionic multiplication and J is required to respect it, then J belongs to a restricted admissible family rather than the full space of metric-compatible complex structures.

This assumption belongs to Mathematical Extension, not to the functional Core.

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21. No-Go Ledger

The No-Go Ledger places hard limits on what may be inferred.

NG0 — Real Self-Revision

A system can possess Gate, Memory, Residual, and Self-Revision while remaining entirely real-valued.

Therefore:

self-revision ⇏ complex structure. (21.1)

The source explicitly identifies this as a valuable negative result. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

NG1 — Persistence

Persistence ⇏ complex structure. (21.2)

NG2 — Recursive Adaptation

Recursive adaptation ⇏ $J^2 = -I$. (21.3)

NG3 — Vector-Space Equivalence

$\mathbb{H} \cong \mathbb{C}^2$ ⇏ unique J. (21.4)

NG4 — Metric-Compatible J

$J^2 = -I$ + metric compatibility ⇏ quaternionically admissible J. (21.5)

NG5 — Circular Phase

S^1 ⇏ four-state coarse graining. (21.6)

NG6 — SU(2)

SU(2) ⇏ nine sectors. (21.7)

NG7 — Dimensional Coincidence

$\dim_{\mathbb{R}}(\mathbb{O}) = 8$ ⇏ eightfold symbolic classification. (21.8)

NG8 — Goal

Goal / reward ⇏ Purpose Belt. (21.9)

These No-Go results are explicitly preserved in the source programme. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

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22. Minimal Observer Criterion

A system should not qualify as a self-revising world-forming observer merely because it maps input to output.

A minimal candidate observer must support:

Observe → Gate → Trace → Historical Retention → Residual Response. (22.1)

A self-revising observer additionally requires:

Residual → Revision. (22.2)

A Purpose-bearing self-revising observer further requires:

Reference ≠ Realisation. (22.3)

and must be able to preserve this distinction through revision.

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23. Minimal World Criterion

An operational world W_D should satisfy at least the following candidate conditions.

W1 — Distinguishability

Some operational states or relations are distinguishable under D.

W2 — Transition Structure

There exist admissible nontrivial transitions.

$x \to x'$. (23.1)

W3 — Gateability

Some candidate outcomes can become committed.

W4 — Traceability

Committed outcomes can persist as historical evidence.

W5 — Residuality

Not every discrepancy is automatically absorbed as confirmation.

W6 — Revisability

Persistent mismatch can alter the observer's policy, interpretation, world model, Purpose, or Declaration.

Thus:

world ≠ state space alone. (23.2)

A stronger operational definition is:

world = state space + admissible transitions + commitment + history + residual + revision. (23.3)

Equation (23.3) is a proposed construction, not a completed theorem.

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24. Core Invariant Candidates

The following are theorem targets rather than established results.

I1 — Trace Preservation [H]

A valid high-level revision should not arbitrarily erase historical evidence that motivated the revision.

I2 — Residual Honesty [H]

The system should not redefine its error measure merely to convert every failure into confirmation.

I3 — Purpose Continuity [H]

Small local evidence should not create arbitrarily large Purpose changes.

A local candidate bound is:

$d_P(P_{n+1}, P_n) \leq K\|R_n\|$. (24.1)

The existence and form of such a bound remain open.

I4 — Revision Accountability [H]

A high-level revision should leave sufficient trace to reconstruct:

previous state, (24.2)

new state, (24.3)

triggering residual, (24.4)

selected revision level, (24.5)

and revision cost. (24.6)

Without this, self-revision can degenerate into retrospective rewriting.

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25. Complex Geometry Is Not Core

The functional architecture does not require complex numbers.

The source development explicitly concludes that Purpose Belt does not need complex geometry to exist and that complex geometry must earn its place by emerging from the minimal Purpose–Observer kernel. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

Therefore:

$J, \omega, \mathbb{C}^2, \mathbb{H}, \mathbb{O}, G_2/SO(4) \notin$ Primitive Core. (25.1)

They belong to optional Mathematical Extensions.

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26. Geometry Entrance Test

Before introducing symplectic or complex geometry, the theory should identify a functionally irreducible structure that motivates it.

The current candidate entrance test compares two update directions.

Let:

U_T = observation / trace update, (26.1)

U_P = Purpose interpretation / revision update. (26.2)

Test:

$U_TU_P \stackrel{?}{=} U_PU_T$. (26.3)

If:

$U_TU_P \approx U_PU_T$ (26.4)

across relevant regimes, then much of the proposed conjugate or symplectic structure lacks necessity.

If robust noncommutation appears:

$[U_T,U_P] = U_TU_P - U_PU_T \neq 0$, (26.5)

then one may investigate whether an antisymmetric infinitesimal structure can be extracted.

The source explicitly proposes this as the correct entrance test for deeper geometry. $\mathbb{O} \to$

27. Optional Purpose Geometry

Conjecture PG-1 [H]

Suppose the Purpose state space carries a positive metric:

$g_P > 0, \quad (27.1)$

and an independently justified nondegenerate antisymmetric form:

$\omega_P. \quad (27.2)$

Define:

$A_P = g_P^{-1}\omega_P. \quad (27.3)$

If:

$-A_P^2 > 0, \quad (27.4)$

define:

$S_P = (-A_P^2)^{1/2}. \quad (27.5)$

Then:

$J_P = A_P S_P^{-1}. \quad (27.6)$

Under suitable compatibility conditions:

$J_P^2 = -I. \quad (27.7)$

This would yield a derived complex operational structure.

However, the crucial unresolved problem is earlier in the chain:

> Does the minimal Purpose–Observer architecture naturally produce the required nondegenerate ω_P?

Until that question is answered, complexification remains a hypothesis.

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28. Optional Structural Declaration Geometry

A separate mathematical extension begins from octonionic geometry.

Let $\mathbb{O}$ be the octonions.

The family of quaternionic subalgebras can be represented by a moduli space related to:

$G_2/SO(4). \quad (28.1)$

A point A in this space selects an associative quaternionic subalgebra:

$\mathbb{H}_A \subset \mathbb{O}$. (28.2)

This motivates a candidate structural declaration:

$D_A: \mathbb{O} \to \mathbb{H}_A$. (28.3)

The interpretation is:

larger relation carrier → selected associative operational world. (28.4)

This is a Mathematical Extension, not a primitive ontology.

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29. Structural Declaration and Operational Polarisation

The two transformations must remain distinct.

First:

8 real dimensions → 4 real dimensions. (29.1)

Under the quaternionic extension, this is interpreted as **Structural Declaration**.

Second:

4 real dimensions → 2 complex dimensions. (29.2)

This is not another dimensional reduction.

It requires an independently selected complex structure J.

Thus:

Structural selection ≠ operational polarisation. (29.3)

The source explicitly separates these two problems and warns against treating $\mathbb{O} \to \mathbb{H} \to \mathbb{C}^2$ as a simple chain of projections. $\mathbb{O} \to G_2/SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1...

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30. Quaternionic Compatibility

Even if a metric-compatible complex structure J exists, it need not preserve quaternionic multiplication.

If $\mathbb{H}_A$ is fixed, a quaternionically admissible candidate may have the form:

$J_{A,u} = L_u$, (30.1)

where:

$u \in \mathrm{Im}\mathbb{H}_A$, (30.2)

and:

$\|u\| = 1$. (30.3)

Then:

$u \in S^2_A$. (30.4)

A combined structural-purpose declaration may therefore be represented as:

$\mathcal{D} = (A,u)$. (30.5)

This yields:

$\mathbb{O} \to \mathbb{H}_A \to (\mathbb{H}_A, J_A, u) \cong \mathbb{C}^2$. (30.6)

The source treats this as a candidate coupling between Structural Declaration and Purpose polarisation, not as an established necessity.

The conceptual relation is:

Structure constrains Purpose. (30.7)

Purpose polarises Structure. (30.8)

This remains outside the Formal Core proper.

---

31. Structural and Purpose Residuals

Under the optional coupled extension, residual may separate into:

R_S = structural residual, (31.1)

and:

R_P = Purpose-polarisation residual. (31.2)

R_S asks:

> Is the declared world grammar itself wrong?

R_P asks:

> Is the world grammar adequate but its current Purpose orientation unsuitable?

This gives two revision scales:

$u_n \to u_{n+1}$ for Purpose revision, (31.3)

$A_n \to A_{n+1}$ for Structural Declaration revision. (31.4)

The distinction is a hypothesis to be experimentally evaluated.

---

32. Revision Timescales

A general architecture may involve several characteristic timescales.

Let:

t = ordinary state dynamics, (32.1)

τ_P = Purpose / interpretation revision, (32.2)

τ_D = structural Declaration revision. (32.3)

A candidate regime is:

$t \ll \tau_P \ll \tau_D$. (32.4)

Equation (32.4) is not a theorem.

It is a testable working hypothesis expressing progressively stronger persistence at higher revision levels.

---

33. Explicitly Superseded Claims

The Formal Core isolates several earlier formulations that should not be retained as established claims.

S1

A simple identification of two different four-dimensional structures with two traditional systems.

Status: Superseded.

S2

Octonionic dimension eight directly derives an eightfold symbolic classification.

Status: Analogy only.

S3

A four-state symbolic system proves complex structure.

Status: Reverse causation.

S4

SU(2) naturally forces nine sectors.

Status: Not established.

S5

Purpose necessarily generates J.

Status: Too strong.

S6

Dirac structure has already been derived from Purpose.

Status: Too strong.

S7

Two complex channels automatically equal a particular semantic Plan/Action or Action/Ledger interpretation.

Status: Semantic hypothesis only.

The source explicitly lists these kinds of claims for downgrade or removal. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

---

34. Formal Falsification Conditions

The theory must specify how its own components can lose status.

F1 — Purpose Reduction

If a simpler utility/world-model architecture reproduces both action and revision behaviour under ontology shifts:

strong Purpose Belt irreducibility fails. (34.1)

The source explicitly identifies this as a decisive failure condition. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

F2 — Attribution Redundancy

If generic residual-driven updating performs equivalently to explicit revision-level attribution across heterogeneous failure causes:

Attribution should be removed from the Core. (34.2)

F3 — Latching Redundancy

If removing level-dependent latching does not produce characteristic instability or drift:

Latching should be downgraded from primitive status. (34.3)

F4 — Dual Residual Redundancy

If residual magnitude alone matches the predictive and revision performance of magnitude plus

direction:

Directional Residual should remain optional. (34.4)

F5 — Complexification Failure

If trace and Purpose updates commute robustly:

$[U_T, U_P] \approx 0$, (34.5)

then the current symplectic/complex route loses its principal functional motivation.

F6 — Quaternionic Extension Failure

If $G_2/SO(4)$ declaration geometry adds no measurable value in compression, prediction, intervention, or model transition:

the quaternionic declaration extension remains mathematical analogy rather than operational theory. (34.6)

---

35. Formal Minimality Principle

The strongest general principle of the Formal Core is:

> **No structure is retained because it makes the theory richer. A structure is retained because removing it destroys a behaviour, distinction, prediction, or derivation that cannot be recovered more simply.**

Let complexity be $C(M)$ and behavioural adequacy be $Q(M)$.

For two models M_1 and M_2:

if $Q(M_1) \approx Q(M_2)$ and $C(M_1) < C(M_2)$, prefer M_1. (35.1)

This is not merely an engineering preference.

It is part of the epistemic discipline of World-Formation Theory.

---

36. The Core in One Operator

For compact representation, define a world-formation episode operator c:

$\Omega_{n+1} = c(\Omega_n; \Sigma)$. (36.1)

A candidate factorisation is:

$c = U \circ A \circ E \circ F \circ T \circ G \circ \Phi$. (36.2)

Read from right to left:

Dynamics → Gate → Trace → Filtration → Residual → Attribution → Revision. (36.3)

The operator becomes recursively world-forming only because the resulting history can change future Declaration and Purpose:

$(D_n,P_n) \rightarrow \text{history} \rightarrow (D_{n+1},P_{n+1}).$ (36.4)

Equation (36.2) is a compact construction, not a claim of uniqueness.

---

37. The Formal Core Thesis

The theory can now be reduced to one substantive statement.

> **A bounded system requires more than states and transition rules if it is to possess an operational world with governed commitment, historical consequence, persistent reference, residual mismatch, and self-revision.**

The minimal world-forming cycle is:

Declaration → Gate → Trace → Filtration → Residual → Attribution → Latching / Revision → Declaration. (37.1)

Purpose adds:

persistent counterfactual reference ≠ realised history. (37.2)

Together they permit a system to distinguish:

what exists in its current representation, (37.3)

what actually happened, (37.4)

what failed to fit, (37.5)

what should change, (37.6)

and what should remain invariant through change. (37.7)

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38. Formal Research Agenda

The Formal Core leaves six primary research problems.

R1 — Minimality

Which of Gate, Trace, Filtration, Residual, Attribution, Latching, Purpose Identity, and Purpose Interpretation are behaviourally irreducible?

R2 — Revision Hierarchy

Can state, policy, world-model, interpretation, Purpose, and Declaration revision be operationally distinguished by different failure signatures and optimal switching costs?

R3 — Residual Geometry

Does separating residual magnitude from residual direction improve diagnosis of noise versus structural mismatch?

R4 — Purpose Geometry

Do trace update and Purpose update exhibit robust, functionally necessary noncommutation?

R5 — Complexification

If a positive metric g_P and antisymmetric structure ω_P emerge independently, does their polar construction produce a useful and stable complex structure J_P?

R6 — Structural Declaration Geometry

Does $G_2/SO(4)$ provide predictive or computational value as a declaration moduli space rather than merely a mathematically attractive analogy?

---

39. Relationship to the Experimental Programme

The Formal Core does not determine which of its candidate components deserve permanent status.

That decision is delegated to experiment.

The next document therefore asks:

Gate → does commitment improve? (39.1)

Trace → does historical consistency improve? (39.2)

Residual → does diagnosis improve? (39.3)

Latching → does stability improve without rigidity? (39.4)

Purpose → does long-horizon coherence improve? (39.5)

Attribution → does revision-level selection improve? (39.6)

Meta-Declaration → does recovery from ontology failure improve? (39.7)

Only after these functional arrows survive should deeper geometry become a primary experimental target.

---

40. Conclusion

World-Formation Formal Core v1.0 is intentionally small.

It does not require:

$\mathbb{O}$, (40.1)

$G_2/SO(4)$, (40.2)

$\mathbb{H}$, (40.3)

$\mathbb{C}^2$, (40.4)

symplectic geometry, (40.5)

or any particular interpretive tradition. (40.6)

Those may later become valuable.

They are not needed to state the foundational problem.

The Core asks only:

> **How can a bounded observer form a world, allow that world to acquire history, detect when its declaration no longer absorbs experience, and revise the very structure through which its history became meaningful?**

Its answer is not yet a completed theorem.

It is a minimal research architecture:

Declaration → Gate → Trace → Filtration → Residual → Attribution → Latching → Revision. (40.7)

Purpose contributes a persistent counterfactual reference:

Purpose Identity ≠ Current Interpretation ≠ Realised History. (40.8)

The programme becomes scientifically meaningful only if these distinctions survive formal reduction and controlled ablation.

The governing principles are therefore:

> **Architecture must earn its complexity.**
>
> **Geometry must earn its necessity.**
>
> **Every important arrow must be allowed to fail.**

The next document is **World-Formation Experimental Programme v1.0**, which converts this Formal Core into a staged sequence of falsifiable experiments.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics,

physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.